Grailcard — Competitive Accuracy Report

Head-to-head evidence: Grailcard vs CardGrader.AI vs Ximilar, anchored to PSA-certified ground truth · August 20, 2026

1. Method — and its honest limits

Five cards were run through multiple systems between August 18–20, 2026: the same images, submitted to **Grailcard** (this product), **CardGrader.AI** (their web app and their v1 Agent API), and **Ximilar card-grader v2** (trained-CV grading API). Two of the five cards are **PSA-certified** — their slab labels provide machine-verifiable ground truth, the rarest commodity in this market. The test set is adversarial by construction: a digital render, a non-database game card, a Japanese-text foil, and two slabbed cards — chosen because easy cards don't separate systems.

Limits, stated plainly: n=5, self-run, and the set was selected to stress systems (all vendors, including us, do better on easy frontal photos of common cards). This is a case-study comparison, not a population statistic. It is, however, fully reproducible: every input image, output, and API response is preserved in the repository, which is more than can be said for any competitor's self-reported "87–95% accuracy," none of which is accompanied by inputs, outputs, or ground truth.

2. The evidence — five cards, all systems

Card (truth)	Grailcard (current build)	CardGrader.AI	Ximilar
Base Set Charizard #4, Unlimited — pristine digital render	**ID correct** (base1-4, 100% w/ visual check) · flagged **"possible digital image"** · grade 9.0 w/ band · sane raw $860 + PSA 8/9 sold medians ($705/$1,450, set-verified)	**Misidentified as 1st Edition** (a ~300× value error) · graded a render CG 9.3 with fabricated corner narrative & back analysis (no back existed) · **PSA 9 $67,000 > PSA 10 $10,000** in one table · used comps it admitted were counterfeit-polluted	Graded the render 7.5 "Near Mint" — no digital-source awareness · centering read 44/56 on a perfectly centered image
Marvel Top Trumps Spider-Man — no database exists for this card anywhere	**"Spider-Man — Top Trumps: Marvel Comics"** via AI vision, labeled not-catalog-verified · graded with scratch marks boxed · valuation honestly absent, eBay sold links offered	Name correct · **fabricated back centering 9.6 with no back photo** · showed $4 raw / $631 PSA 10 while its own analysis text admitted "NO SOLD COMPS FOUND … all value fields $0.00"	not run (credits)
Mega Charizard X ex (MEP promo) — Japanese art lettering, English rules	**"Mega Charizard X ex" 85%** after ambiguity fixes (an earlier build hit the generic-name trap — found & fixed same day) · honest full-art n/a's	not run	not run
Monkey.D.Luffy OP07-109 SR — PSA-certified GEM MT 10 (cert 96811063, verified on PSA's registry)	**Slab detected: "PSA GEMMT 10", cert read exactly, registry link** · card ID exact via printed set code · declares the certified grade authoritative instead of re-grading plastic	**Estimated 9.5** (−0.5 vs certificate) — **did not notice the PSA slab it drew boxes around** · set ID correct · coherent eBay comps this time	**Graded 6.0** (−4.0 vs certificate) — honestly scored slab glare, but no slab awareness
Base Set 1st Ed Charizard — PSA-certified GEM MT 10 (cert 25201088), tiny 250px photo	**Slab detected: "PSA GEM MT 10"** (cert one digit off at this resolution — registry link enables human check) · refused a real grade (photo too small), provisional impression only, **no grading recommendation issued** · card-set ID missed when the LLM tier was down (known gap)	**ID correct** ("1999 Base Set 1st Edition Charizard Holo 4/102" — via their API, identify module) · **no slab/cert awareness in the response**	not run (credits)

3. Scorecard and accuracy claims

Metric (this test set)	Grailcard	CardGrader.AI	Ximilar
Certified-grade agreement (2 PSA-slabbed cards)	**2/2 exact — reads the label, links the cert**	**0/1 exact (9.5 vs 10); slab never detected**	**0/1 (6.0 vs 10); slab never detected**
Card identification fully correct	**4/5 (miss: slab-card set when LLM tier was down)**	**3/5 (1st-Edition hallucination; "2003 Donruss Legends" fabrication w/ wrong-player comps on a 6th test card)**	n/a (not an identifier in these runs)
Fabricated data in final output (grades/values presented for things not measured or not existing)	**0 — unmeasurable = "not assessable"; no data = "no data"**	**4 instances (render sub-grades, phantom back centering, $631 for a card with zero comps, contradictory PSA 9>10 pricing)**	**0 — grades only what it sees**
Digital-source (render/screen) awareness	**flags it**	**none observed**	**none observed**
Refuses / caveats bad input	**quality gate + provisional labeling + no money verdicts from rejected photos**	**always answers**	**grades whatever it receives**

> **The accuracy claim we are prepared to make (and defend with the artifacts):**
> On a five-card adversarial test set containing two PSA-certified ground truths, Grailcard was the **only system that reported the certified grade correctly (2/2)** — because it is the only system that notices a card is already graded. It achieved the **highest identification accuracy (4/5 vs 3/5)** and was the only "full-report" system with **zero fabricated numbers** in its final outputs. We do not claim a population-level accuracy percentage — no one in this market can, honestly; vendors' self-reported 87–95% figures ship without inputs, outputs, or ground truth. Ours ships with all three, and the planned public prediction-vs-PSA dashboard will turn this into a live, falsifiable statistic.

Fairness notes: CardGrader's Luffy estimate (9.5 vs certified 10) is genuinely good photo-only grading; its slab-Charizard identification was exact; and its eBay comp pipeline is real when comps exist. Ximilar's 6.0 on the slabbed Luffy honestly scored what the photo shows (glare on plastic) — the failure is contextual, not perceptual. Grailcard's own development history includes every same-class error (wrong-card comps, a "Klara" false match, a fabricated-feeling 100%) — the difference is each became a same-day fix with a regression test, preserved in the git history.

4. What "perfection" requires — the complete stack and its cost

Gap that remains	What fixes it	Where	Cost
Photo quality dominates every grade (the #1 accuracy lever, proven repeatedly above)	Guided capture: live brackets refusing tilted/shadowed/small shots before they're taken	Build in-house (browser camera + our quality checks)	**$0** — engineering time only
Trained condition grading (currently on heuristic fallback — credits exhausted)	Ximilar card-grader v2 credits; later, both sides per scan	ximilar.com	**€10** ≈ 90 grades · **€59/mo** ≈ 1,000 · long-term: own models trained on accumulated scans → ~$0
Sports cards (identification + prices) — the biggest coverage hole	PriceCharting/SportsCardsPro DB + graded prices; Ximilar sport_id for photo-ID (same credit pack)	pricecharting.com	**~$6/mo** flat
LLM identifier flakiness (Gemini free-tier 503s caused our only current-build ID miss)	Paid Gemini tier (or Claude API) — same code, reliable	Google AI Studio	**~$5–15/mo** at current volume
Text-independent identification (foreign, art-only, worn cards)	Visual-search index: perceptual-hash/embedding index over ~80k catalog images we already access free	Build in-house (overnight job)	**$0**
In-app eBay sold comps (links are live; parsed averages are not)	eBay Partner Network + Browse API	eBay (application)	**Free** (approval process)
Slab cert auto-verification (we read labels; we don't verify them)	PSA public cert lookup wired to the detected cert number	psacard.com	**Free**
Population reports (PSA-10 scarcity → sharper value math)	Scrydex	scrydex.com	$29/mo (optional)
Production reliability (this all runs on one 7GB laptop that OOMs)	Small VPS, job queue, process supervision, backups	any cloud	**~$10–25/mo**
The moat: a live, public accuracy statistic	Prediction-vs-PSA dashboard fed by user-submitted cert numbers, auto-verified	Build in-house	**$0**

Totals. Perfection-stack first month: **≈ $30–45** (€10 Ximilar + $6 PriceCharting + ~$10 Gemini paid + hosting). Steady state before real volume: **≈ $25–45/mo**. At ~1,000 scans/mo: **≈ $100–150/mo** all-in, ≈ 8–12¢ marginal cost per fully-graded scan against a ~25¢ market price — with the marginal cost trending toward ~1¢ once in-house models replace per-scan API grading. Everything already live today (7 game catalogs, prices, PSA sold data, slab reading, AI identification, measurement pipeline) runs on **$0/mo**.

Artifacts backing this report: all five input images in `samples/` and scan storage; API responses reproduced in the session log; CardGrader agent #58 scans #413258 & #413864; PSA certs 96811063 & 25201088 verifiable at psacard.com/cert. Grailcard git history documents every error found and its fix. Generated August 20, 2026.